Software Systems

Marks: 60

Question 1

(a) Draw an Entity Relationship diagram for the following description of a bank's software system. Show the entities, relationships, cardinalities of relationships, participation types of relationships, attributes, and primary keys.

The bank has several customers, branches, and accounts. Each branch has a unique branch code and a branch name. Each customer has a unique id, name, address, and the total amount across all their accounts. Each account has a unique account number (unique across all branches of the bank), the amount of money currently in the account and an account type. The possible account types are personal, business, joint, salary, savings and current. The same account may be of more than one types, e.g., a personal account may be a joint savings account, etc. An account belongs to a specific branch.

An account may be owned by one or more customers (in case of joint accounts). A customer may own one or more accounts. There cannot be a customer who does not own an account anywhere. Similarly, there cannot be an unowned account.

A customer can deposit money into an account and withdraw money from it. Each deposit and withdrawal has an associated amount (the amount deposited or withdrawn) and a timestamp. A customer can make any number of deposits and withdrawals. Similarly, an account may be deposited into and withdrawn from any number of times.

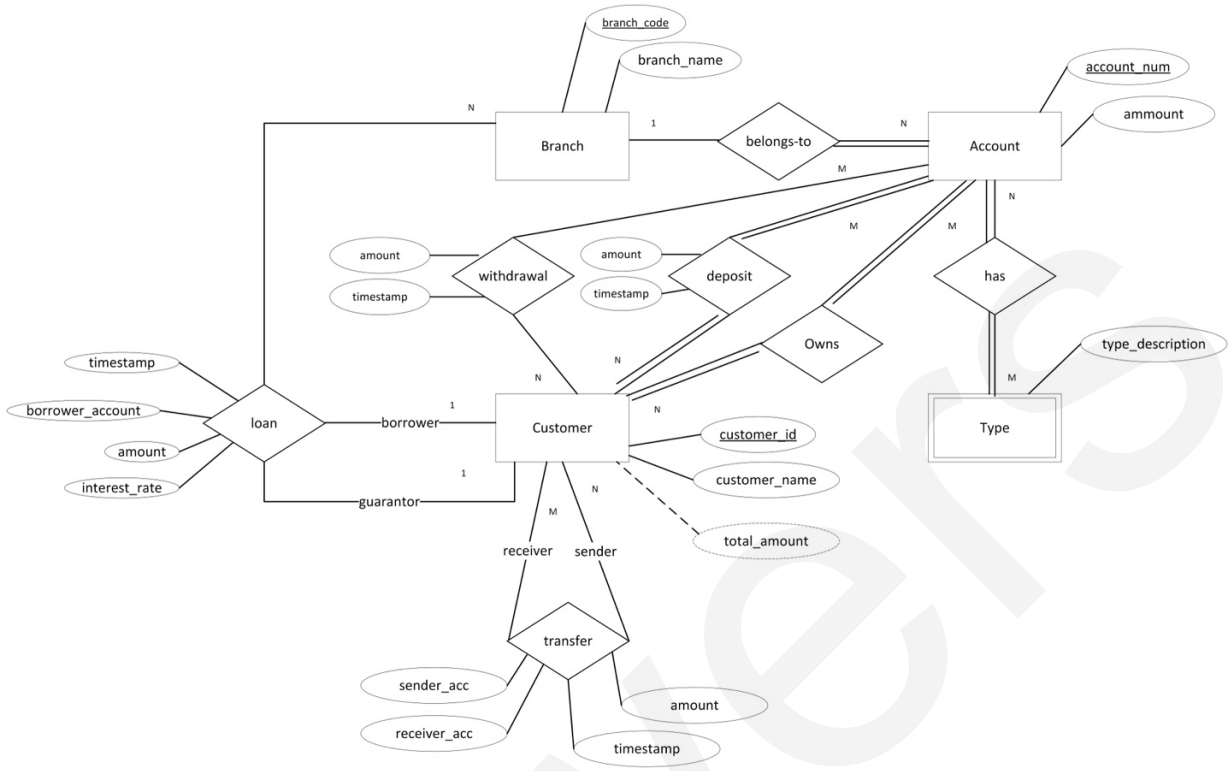
A customer may transfer money to another customer. A transfer has an associated sender and receiver account numbers, the amount of money transferred, and a timestamp. It is possible for a customer to never withdraw money or participate in a transfer. However, each customer must make an initial deposit into each of their accounts. Similarly, an account must be deposited into at least once, however, it is possible that money is never withdrawn from it.

A branch can give a loan to a customer. When giving the loan, another customer must act as the guarantor. A loan has an associated amount, interest rate, borrower's account number, and timestamp. A customer may borrow no more than one loan at a time. Similarly, a customer may not act as the guarantor in more than one loans at a time. A branch can hand out any number of loans.

[10]

Answers

E-R Diagram



(b) Convert the Entity Relationship model from (a) into a relational schema. Highlight the primary key of each relation in the schema.

[5]

Customer(customer_id, customer_name, customer_address)
Branch(branch_code, bname)
Account(account_num, amount, branch_code)
Owns(customer_id, account_num)
HasType(account_num, type_description)
Deposit(customer_id, account_num, timestamp, dep_amount)
Withdrawal(customer_id, account_num, timestamp, wd_amount)
Transfer(sender_id, receiver_id, timestamp, sender_acc, receiver_acc)
Loan(borrower_id, guarantor_id, branch_code , timestamp, amount, interest_rate)

(c) The following relational schema represents a miniature course enrolment system.

```
Student(student_id, student_name, average_grade)
Enrolment(student_id, course_id, date)
Course(course_id, course_title, department, credit_hours)
Teaches(lecturer_id, course_id)
```

Lecturer(lecturer_id, lecturer_name, department)

Write SQL queries to perform the following tasks:

(i) List the ids and names of students who have only enrolled in courses offered by the department of 'Computer Science.' (You may assume that every student has enrolled in at least one course).

```
SELECT student_id, student_name FROM Student
WHERE student_id NOT IN
(
   SELECT student_id FROM
   Student NATURAL JOIN Enrolment NATURAL JOIN Course
   WHERE Course.department!= 'Computer Science'
);
```

(ii) List the ids and names of students who are being taught by three or more different lecturers.

```
SELECT student_id, student_name FROM
(
   SELECT student_id, student_name, COUNT(lecturer_id) as lec_count
   FROM
            (
            SELECT student_id,student_name, lectutrer_id FROM
              Student NATURAL JOIN Enrolment NATURAL JOIN Teaches
            )
   GROUP BY student_id
)
WHERE lec_count >= 3 ;
```

Note: You may use NATURAL JOIN to join tables on columns with identical names, e.g., *Student* NATURAL JOIN *Enrolment* will join the two tables on student_id.

[5 + 5]

Question 2

(a) Answer the following questions related to application and transport layer protocols:

i. Give two different examples of application layer protocols that rely on caching to improve the user perceived performance of the network application. Indicate why caching is used, where the caches are located, and how they work.

Answers

DNS (Domain Name Service)

DNS uses caching by storing host-name to IP mappings at the local end systems, local DNS servers and regional DNS servers to offload root DNS servers.

HTTP (HyperText Transfer Protocol)

HTTP caches copies of popular objects and webpages closer to clients. The caching may be done in the local browser's cache, an ISP cache or web proxy server. It makes use of the Conditional GET to only fetch pages from the original server (instead of the cache) if pages have been modified since the last time they were cached.

ii. Indicate if TCP or UDP would be a better fit for the service requirements of the following applications and briefly explain why. State any assumptions that you make for each application:
– Text based instant messaging app
– Clock synchronization between computers on the internet
– Online multiplayer first-person shooting game

– Text-based instant messaging app:
TCP will be better. This app needs reliable data delivery whereas some delay is tolerable.

– Clock synchronization between computers on the Internet:
UDP is ideal for this app. The message sizes are small, and the app is highly delay-sensitive but loss-tolerant.

– Online multiplayer first-person shooting game:
UDP is better considering fast response (delay sensitive) nature of the game application. On the other hand, the application is also loss tolerant.

(b) Suppose a browser is fetching a webpage. The base HTML file references 13 very small objects on the same server. Assuming: an RTT of 3ms, each object small enough to fit a single TCP segment, no transmission delays, how much time does it take to fetch the complete page, when using:

(i) Non-persistent HTTP with no parallel TCP connections.
(ii) Persistent HTTP with no parallel TCP connections.
(iii) Non-persistent HTTP with 3 parallel connections at max. Note: the connections can only be created in serial.
(iv) Persistent HTPP with no parallel TCP connections, just like in (ii). However, this time we also wish to account for the DNS query. Assume that we need to visit 4 servers (including the local DNS) before the IP address is retrieved. Assume DNS querying to be iterative and the transport layer protocol of DNS to be UDP. All RTTs are 3ms.

[2+2+3+2]

Answers

(i) Before the 13 objects the base HTML file is fetched. This is the file that contains links to the 13 objects. This step takes in total 2RTT (one for the connection the other for the file assuming here that it can be fetched in a single RTT).
After this, for each object there is 1RTT for TCP connection and 1RTT for the object fetch.
Total time: 2RTT + 13 * (2 RTT) = 28 RTT, that is: **84 ms**

(ii) Persistent, no parallel connections:
1RTT + 1RTT + 13 RTT = 15 RTT, i.e., **45 ms**

(iii) Non-persistent with 3 parallel connections:
2RTT for base file with one connection
For the next 12 objects, we need to set up 3 parallel connections four times, which takes 3RTT each time (done in serial), and the 3 objects are fetched in parallel in 1RTT. Therefore, a total of 16RTT for the 12 objects.
For the 13th object, a 2RTT, one for connection setup and one for fetch
Total: 2 + 16 + 2 = 20 RTT = **60 ms**

(iv) Time for web-page fetch same as in (ii): 45 ms
Time for DNS: 4 * 1RTT = 12 ms (no connection time need as using UDP).
Total: **57 ms**

(c) Host X needs to send a message consisting of 9 packets to host Y. The two hosts are using a sliding window Go-Back-N strategy with window size of 3. All packets are available for transmission immediately. Assume that every 6th packet that X transmits gets lost. But no ACKs from Y ever get lost. Also assume that the medium delivers packets in order. What is the number of packets that X will need to transmit in order to send the entire message to B?

[5]

Packets 1-8 are sent by the sender before detecting the first loss (of packet 6)

The windows at the sender side may be imagined as follows:

[1, 2, 3], 3 packets sent immediately.
[2, 3, 4], after ACK 1 is received, 4th packet sent.
[3, 4, 5], after ACK 2 is received, 5th packet sent.
[4, 5, 6], after ACK 3 is received, 6th packet sent.
[5, 6, 7], after ACK 4 is received, 7th packet sent.
[6, 7, 8], after ACK 5 is received, 8th packet sent.

Timeout for packet 6.

Answers

[6, 7, 8] Packets 6, 7 and 8 are resent: total 11 packets sent till now.
[7, 8, 9], after ACK 6 is received, 12th packet sent (this is packet 9, which will be lost).
[8, 9, -], after ACK 7 is received,
[9, -, -], after ACK 8 is received.

Timeout for packet 9.

Packet 9 is resent: total 13 packets sent till now.

[9, -, -], after ACK 9 is received.

The transmission is complete in **13 packets.**

Question 3

(a) Answer the following questions related to Network and Data Link layers:

(i) Suppose a router performing NAT (Network Address Translation) with public address 205.10.221.50 is operating on behalf of the nodes in your home, which are part of the internal private network 192.168.4.0/24. When a client with address 192.168.4.2 sends a TCP packet to an outside server at 201.20.4.7 (as shown in the table below), how does NAT modify the packet?

Original packet (before NAT)

	Source	Destination
IP	192.168.4.2	205.10.221.50
TCP Port	5555	20

After NAT

	Source	Destination
IP	?	?
TCP Port	?	?

(ii) What information (mapping) does the router need to keep internally in order to properly translate and forward the server's response?

[3+3]

(i) NAT modifies the packet as follows:

Original packet (before NAT)

	Source	Destination
IP	192.168.4.2	205.10.221.50

After NAT

	Source	Destination
IP	205.10.221.50	205.10.221.50
TCP Port	PORT1	20

Answers

TCP Port	5555	20

PORT 1 is 16-bit port number (from the unreserved port number space) generated and maintained by the NAT.

(ii) NAT will need to maintain a mapping between (205.10.221.50, PORT1) and (192.168.4.2, 5555) to properly forward the server's response to the correct client process in the private network. Therefore, PORT1 must uniquely identify the (192.168.4.2, 5555) pair, i.e., the client process in the private network.

(b) This part relates to subnetting and IP addressing. Suppose an ISP owns the block of addresses of the form 145.126.252.0/22. The ISP wishes to create three subnets from this block: Subnet 1, 2 and 3. Subnet 1 needs a minimum of 350 interfaces, Subnet 2 needs a minimum of 105 interfaces, and Subnet 3 needs a minimum of 221 interfaces. Provide three network addresses (of the form a.b.c.d/x) for the three subnets that satisfy these constraints. Make sure the address spaces don't overlap.

[4]

We can write the last two octets of the IP address in binary to make things clearer:

145.126.252.0 / 22

= 145.126.11111100.00000000 / 22

The underlined bits are the ones available to the ISP.

We may divide these into three subnets as follows:

Subnet 1: 145.126. 111111**0**x.xxxxxxxx / 23, i.e., 145.126. 252.0 / 23. This allows subnet 1 to use 9 bits, which are sufficient for 350 interfaces.

Subnet 2: 145.126. 11111**10**.xxxxxxxx / 24, i.e., 145.126. 254.0 / 24. This allows subnet 2 to use 8 bits, which are sufficient for 105 interfaces.

Subnet 3: 145.126. 11111**11**.xxxxxxxx / 24, i.e., 145.126. 255.0 / 24. This allows subnet 3 to use 8 bits which are sufficient for 221 interfaces.

Clearly, the three address spaces don't overlap.

(c) The following figure shows a network of hosts, routers and switches.

(i) How many subnets does the network have?

4 subnets

Answers

Nodes H2 to H4 connected to the network before H1, all using DHCP. Finally, H1 connected to the network using DHCP.

(ii) Which node (or nodes) would have received the DHCP request from H1? Which node (or nodes) would have responded to this request?

H2 and R1 receive the request. R1 responds with H1's IP, gateway router IP (in this case R1's own IP), local DNS IP.
(all packets pass through the switch S1, but because it is transparent to this communication, we wouldn't say that it received or responded to them)

Now H1 needs to send an IP datagram to H3 (H1 knows its IP address). However, H1's ARP table is empty, hence it needs to send out an ARP request. The ARP tables at all other nodes are fully populated and up to date.

(iii) Which IP address does H1 send in the ARP query to get its corresponding MAC address?

H1 will query with the IP of R1 (its next hop gateway router), i.e., 192.168.5.1, to get the MAC address of R1.

(iv) Which nodes receive H1's ARP request? Which nodes respond to H1's ARP request?

H2 and R1 receive H1's ARP request. Since both of their ARP tables are fully populated, they both respond to the query.

(v) How many more ARP requests need to be made before the packet reaches its destination H3?

No other ARP requests are needed as all other ARP tables are fully populated.

Answers

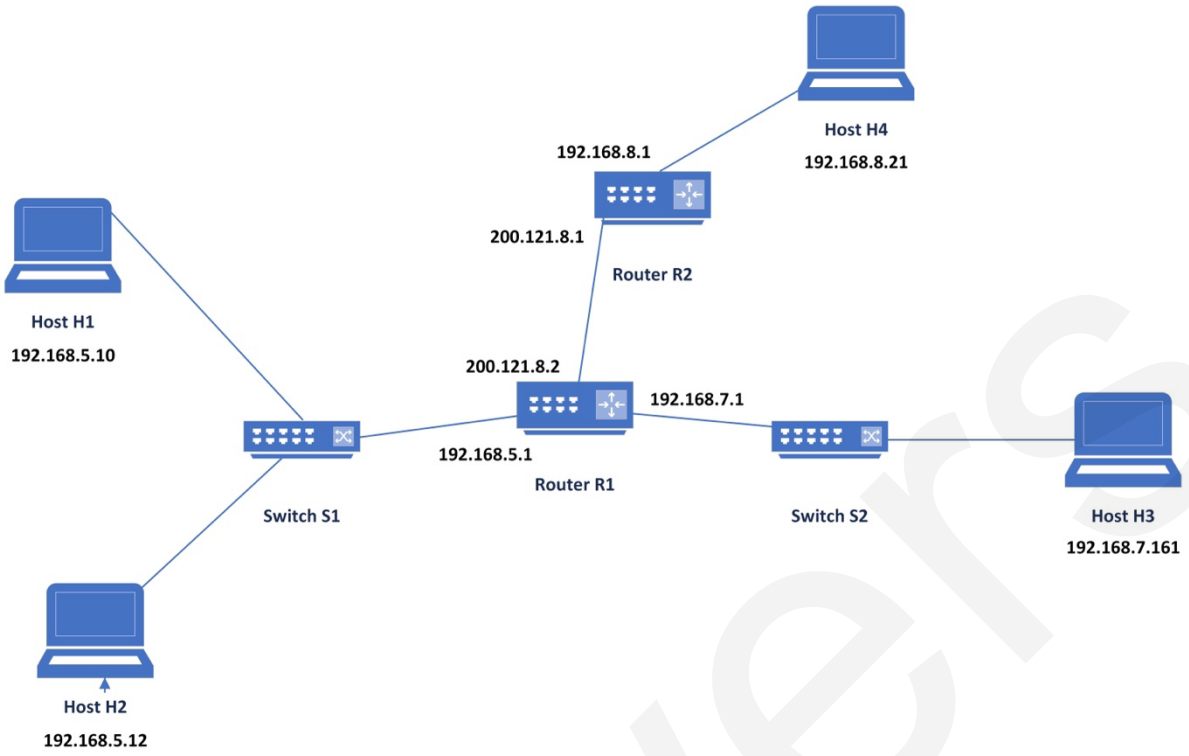


Figure 1

[1+1+1+1+1]

Answers